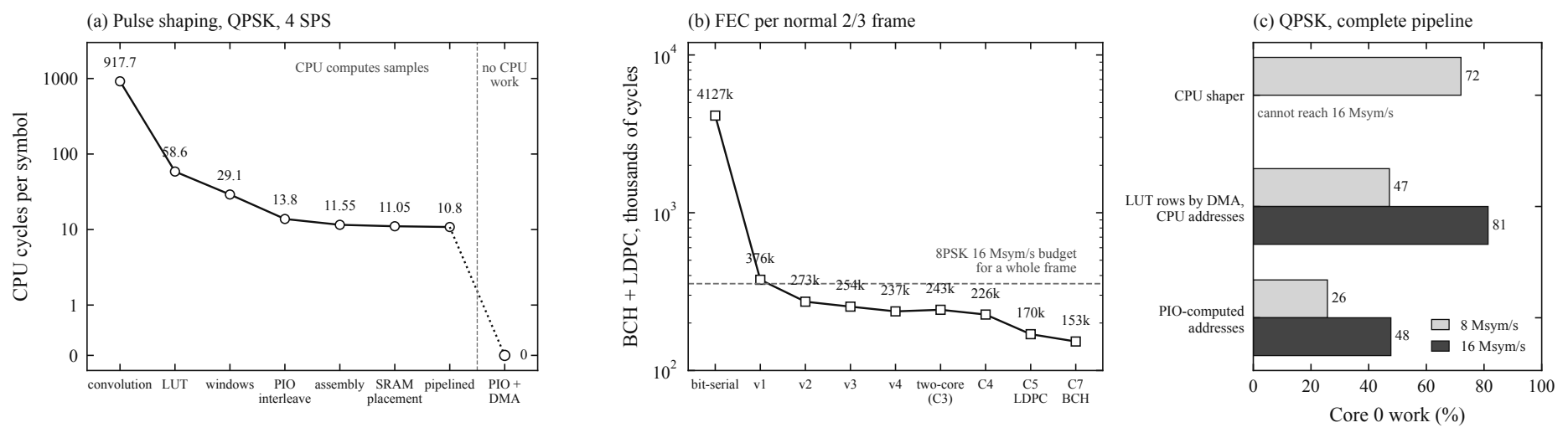
(a) Pulse shaping, QPSK, 4 SPS
CPU cycles per symbol
1000
100
10
1
0
917.7
58.6
29.1
13.8
11.55
11.05
10.8
0
CPU computes samples
no CPU work
convolution
LUT
windows
PIO interleave
assembly
SRAM placement
pipelined
PIO + DMA
(b) FEC per normal 2/3 frame
BCH + LDPC, thousands of cycles
10^4
10^3
10^2
4127k
376k
273k
254k
237k
243k
226k
170k
153k
8PSK 16 Msym/s budget for a whole frame
bit-serial
v1
v2
v3
v4
two-core (C3)
C4
C5 LDPC
C7 BCH
(c) QPSK, complete pipeline
CPU shaper
cannot reach 16 Msym/s
72
LUT rows by DMA, CPU addresses
47
81
PIO-computed addresses
26
48
8 Msym/s
16 Msym/s
0
20
40
60
80
100
Core 0 work (%)